Couchbase

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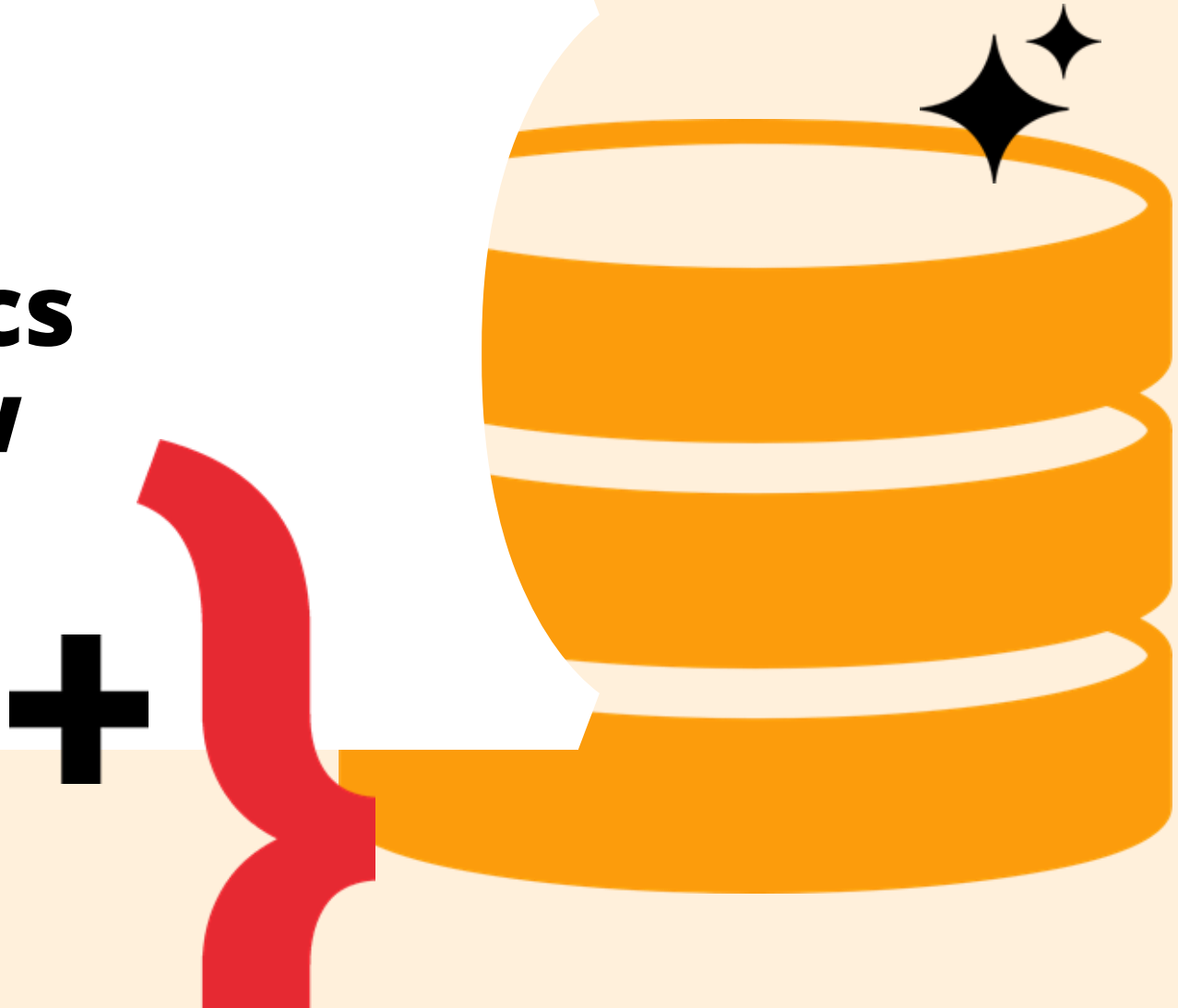
Couchbase Analytics Technical Overview



March 2026

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Couchbase Analytics

Capella Analytics

- Our fully managed analytical database-as-a-service in the cloud
- Formerly *Capella Columnar*

+

Enterprise Analytics

- Same innovation to on-prem and self-managed deployments

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Our Next Generation Analytics

A groundbreaking NoSQL analytics solution, designed for flexible data and application owners

Couchbase Analytics

NoSQL

- Schema Flexibility
- High Availability
- Scalability
- Developer Friendliness

Analytical

- Columnar Storage
- Cost Based Optimizer
- Massive Parallel Processing
- Storage/Compute Separation

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Couchbase Analytics | Key Use Cases

More Personalized User Experiences

Operational Analytics

Embedded Analytics

AI/BI

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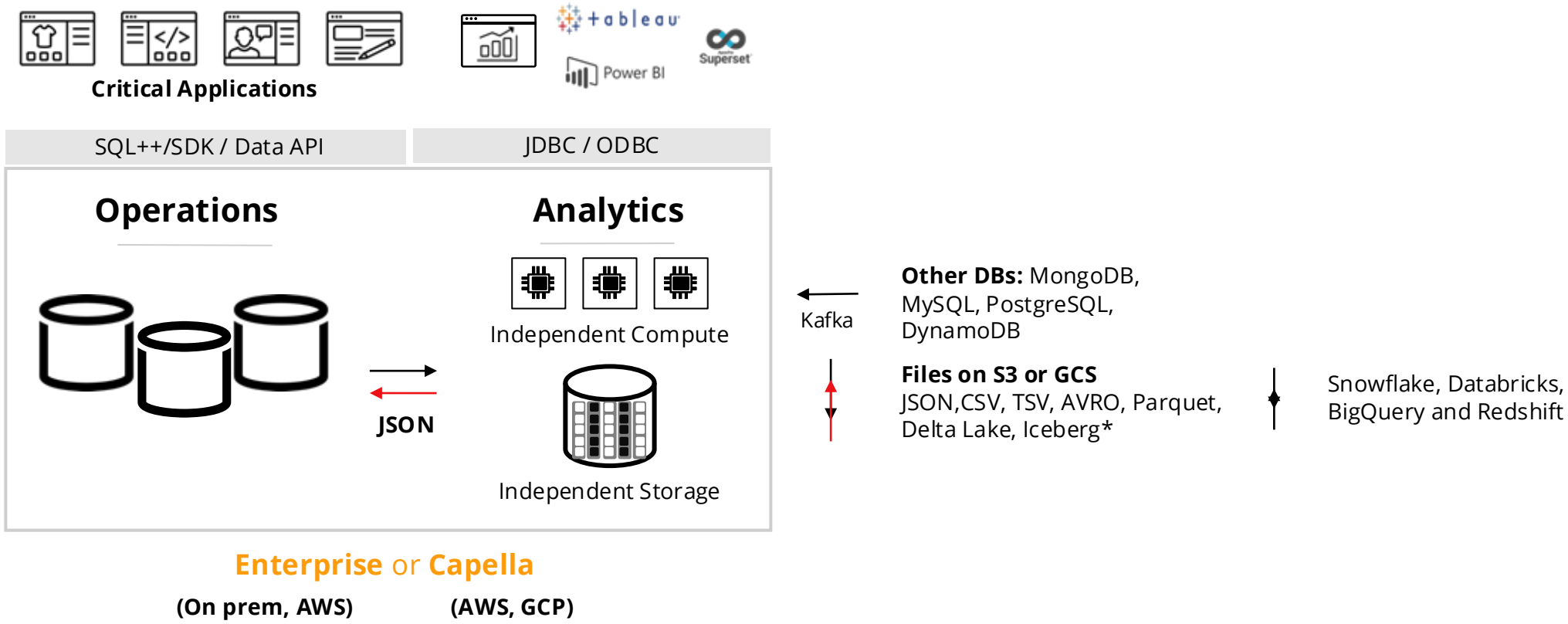
Couchbase Analytics | Architecture



Combines NoSQL flexibility with the performance of modern data warehouses

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(*) coming soon

Couchbase Analytics | Core Pillars

	Capella Analytics	Enterprise Analytics
Architecture	• **Single Control Plane** across multi-cluster • **Data Plane** isolated in a separate VPC within a managed account	• **Self-managed** by customers • Available for both On-Prem and Cloud environments
Deployment	• **Fully Automated** Deployments • **Multiple or Single AZs** on AWS and GCP (Azure coming soon) • **Choice** of Compute Instances • Choice of 2 Support **Plans**	• **Standard Server HW (x86 or ARM)** • **AWS S3** and **NetApp StorageGRID** supported • In future: Azure Blob, AWS Outpost, GCS for Self-managed + Dell EMC ECS, IBM Storage, Minio etc for On-prem.
Database Engine	• **Native Columnar** Storage for JSON • Compute/Storage **Separation** • Massively Parallel Processing • SQL++ and Cost Based Optimization	***Same as Capella Analytics***
Integration	• **Zero ETL** to integrate **multiple sources** (MongoDB, MySQL, Postgresql, DynamoDB) with Bring your own Kafka • Query **External Data Collections** • **Tableau , PowerBI** and **SuperSet** Support • **Write Back** to S3, GCS, Capella database or Couchbase Server	***Same as Capella Analytics***
Operations	• Organize Clusters in **Projects** and **Teams** from SSO groups • Monitoring UI and Query Analysis; Turn Cluster **on/off** • **On demand Scaling** Compute; Automated **Backup/Restore** to Cloud	• Node Management: Add/Remove Node, Join existing cluster, Failover • Monitoring Node resources and Performance Metrics via Web console, CLI and the REST API
Security	• Single Sign-On (SSO) & Role-Based Access Control (RBAC) • **VPC peering, Private Links** & Always **Encrypted** at Rest, In-flight • Designed to comply with GDPR, SOC II, HIPAA, PCI DSS & CSA STAR	• Single Sign-On (SSO) & Role-Based Access Control (RBAC) • Node-to-Node Encryption

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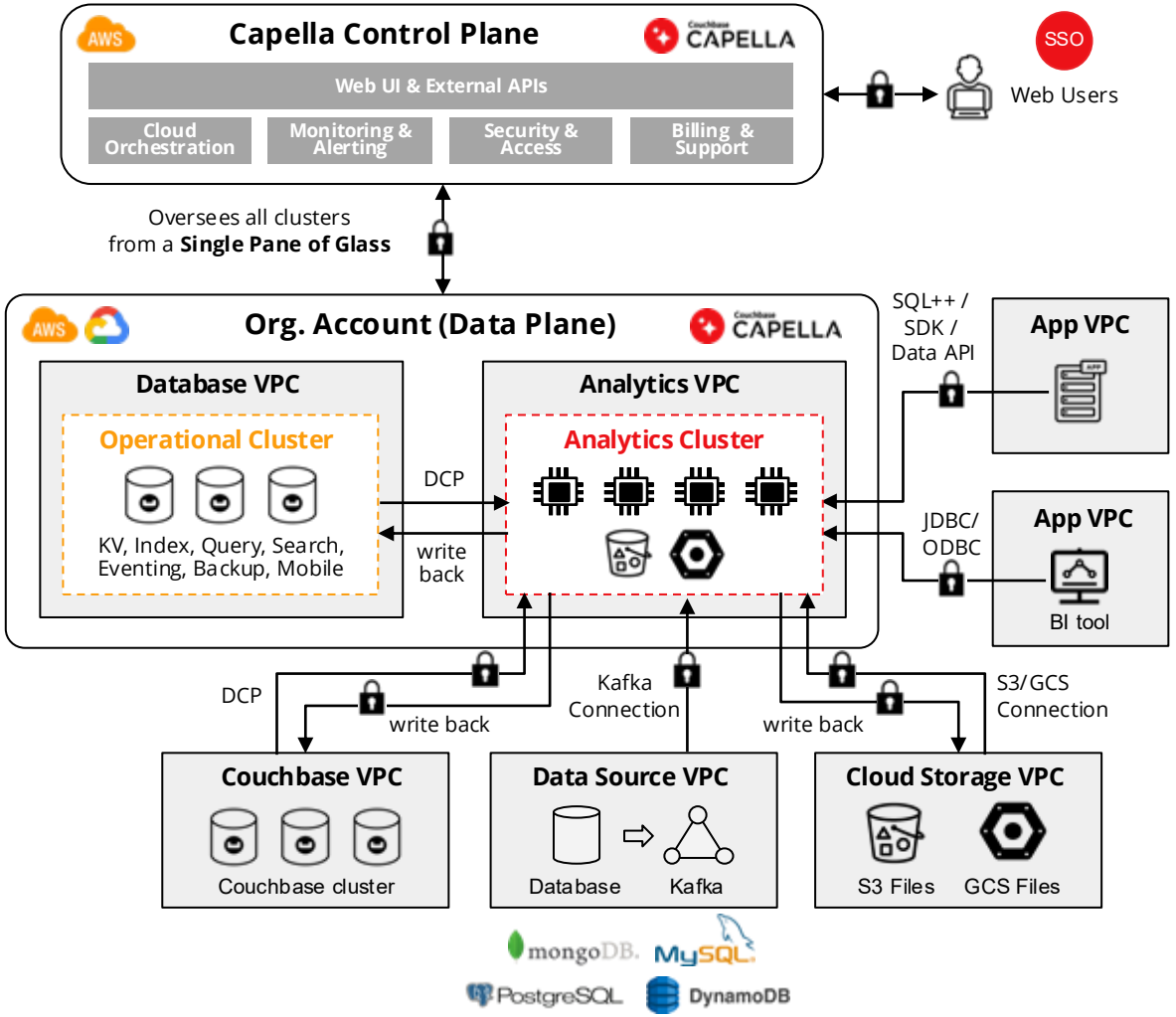
Architecture & Deployment

Fully Managed and Self Managed, Secure and Highly Scalable & Available

Architecture | Capella Analytics Overview



Capella Control Plane

- Manages the Cloud Orchestration, Monitoring & Alerting, Security & Access, Billing & Support
- Is the Access Point for Organization Web UI Users
- Single Control Plane across multi-cluster

Organization Account (Data Plane)

- Account Isolation: 1 Account per Organization
- Cluster Isolation: 1 VPC per Couchbase cluster
- AWS and GCP support - Azure in future
- Compute and Storage separation
- Clusters deployed within a region, across AZs
- Full management of Kafka and S3/GCS Connections

Applications

- Connect directly to Clusters
- Multiple connectivity options: over Public Connection, through VPC Peering or Private Link
- All communications are encrypted

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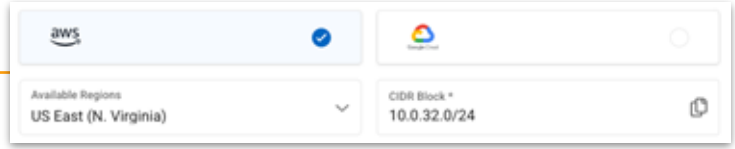
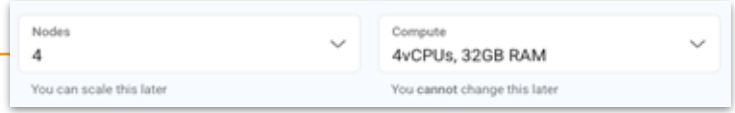
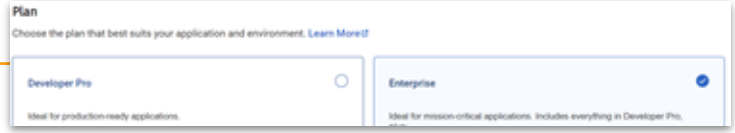
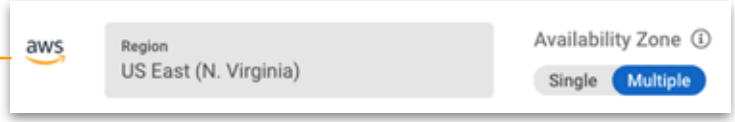
Deployment | Capella Analytics Cluster

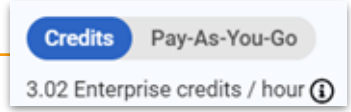
Cloud	Choose Cloud Provider and Region • AWS and GCP (Azure in the future)
Configuration	Chose number of Nodes and Compute type • Number of nodes can be changed post-deployment • Compute type can NOT be changed post-deployment
Support Plan	Choose Developer Pro or Enterprise • Plan defines capabilities, backup and SLA
Availability	Choose the number of Availability Zones • Single or Multiple (two) Availability Zones
Billing Option	Choose Credits or Pay-as-you-Go • Estimated cost available • Excludes storage, variable backup and data transfer costs
Deploy	Couchbase Best Practices automatically enforced • Dedicated storage bucket automatically assigned

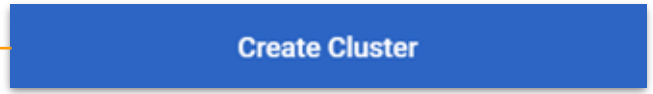












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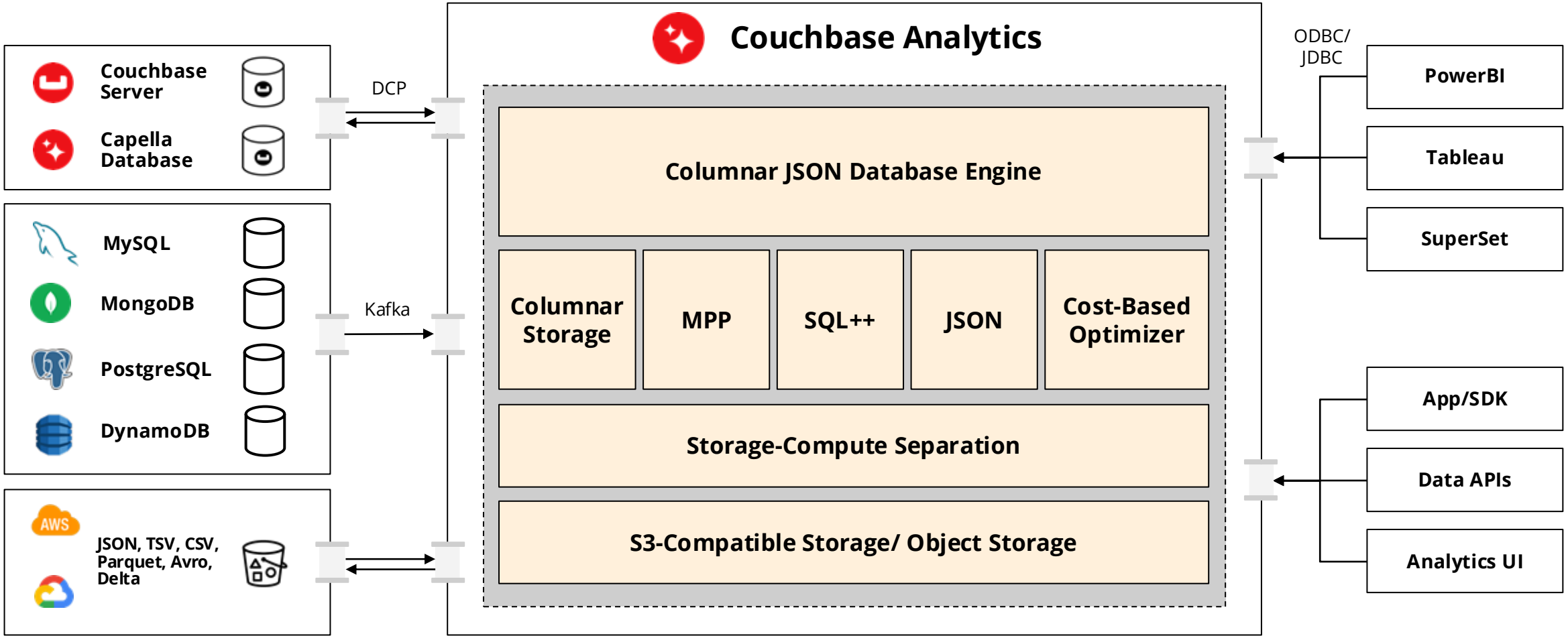
Architecture | Under The Hood



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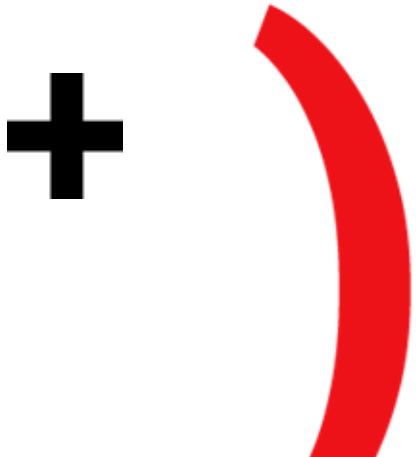
Analytics JSON Database Engine

Completely Architected For JSON Analytics



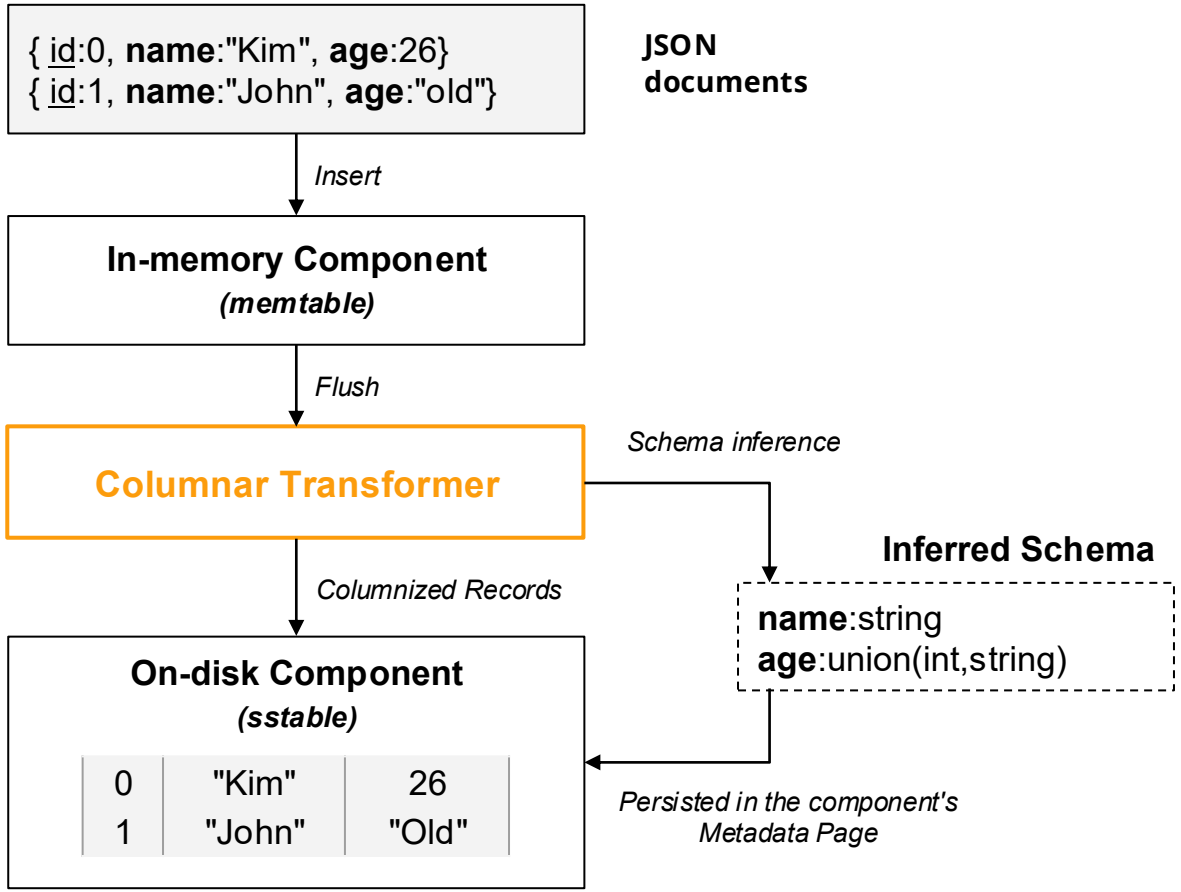
Analytics Storage | Columnization of JSON Data

Motivations

- Column stores group values from the same field for faster analytical queries
- Couchbase Analytics brings columnar storage to NoSQL and JSON
- Boosts query speed while preserving JSON's flexibility

Approach

- JSON documents are first inserted into memory
- Columnar Transformer infers schema when data is flushed to disk
- Data stored in a compact, efficient binary columnar format
- Schema stored separately from data to reduce size
- Columns compressed and encoded to reduce I/O



Data Ingestion Workflow

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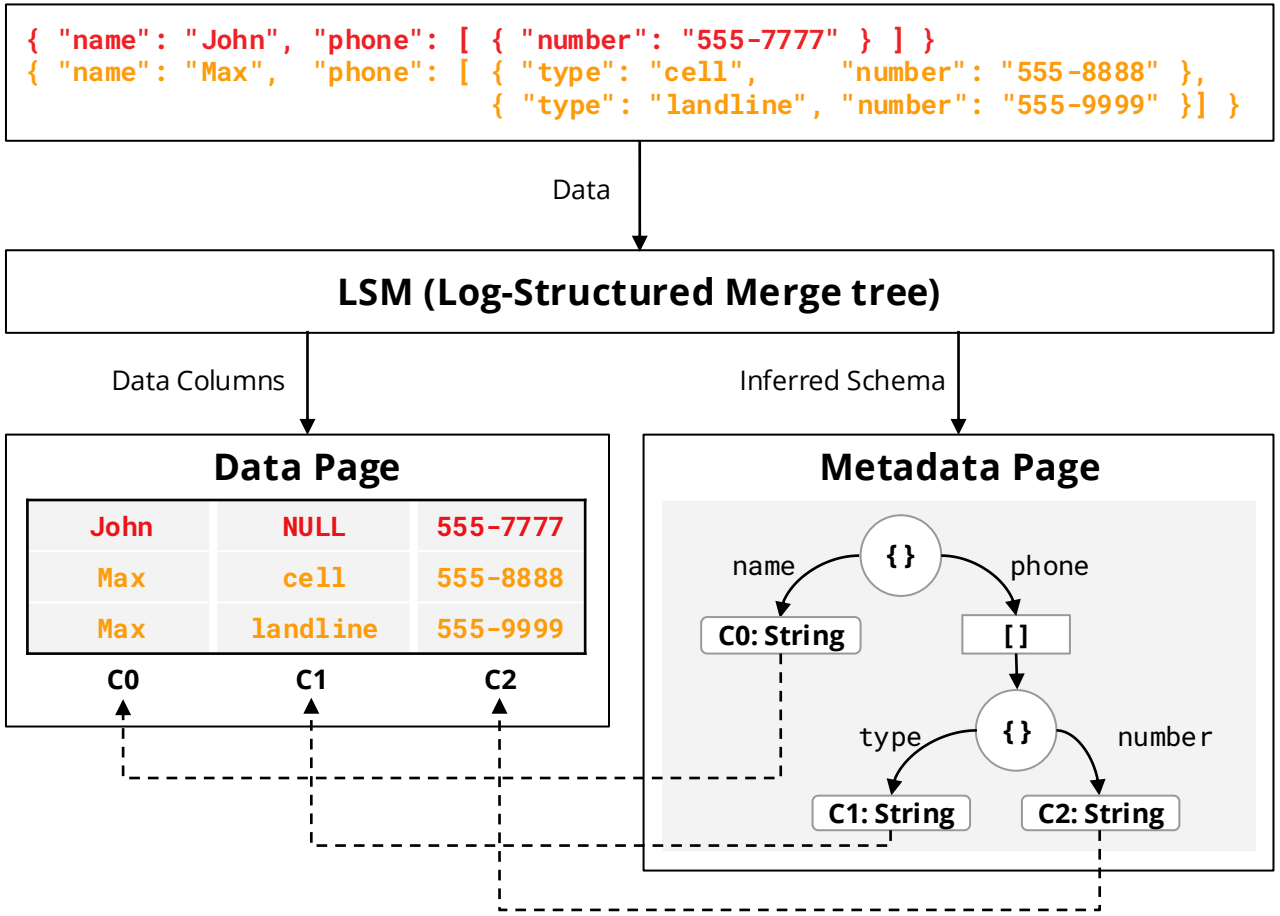
Analytics Storage | Columnization of JSON Data *(Cont'd)*

Column-wise Store

- Built on top of an LSM (Log-Structured Merge) engine
- No need for data to fit in memory
- Custom columnar format—more flexible than Parquet
- Data and schema stored in separate LSM components

Learn More

- See *Keep Calm and Column Wise* blog



Data Columns and **Inferred Schema** are stored separately in each LSM component

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Cost Based Optimization | For SQL++

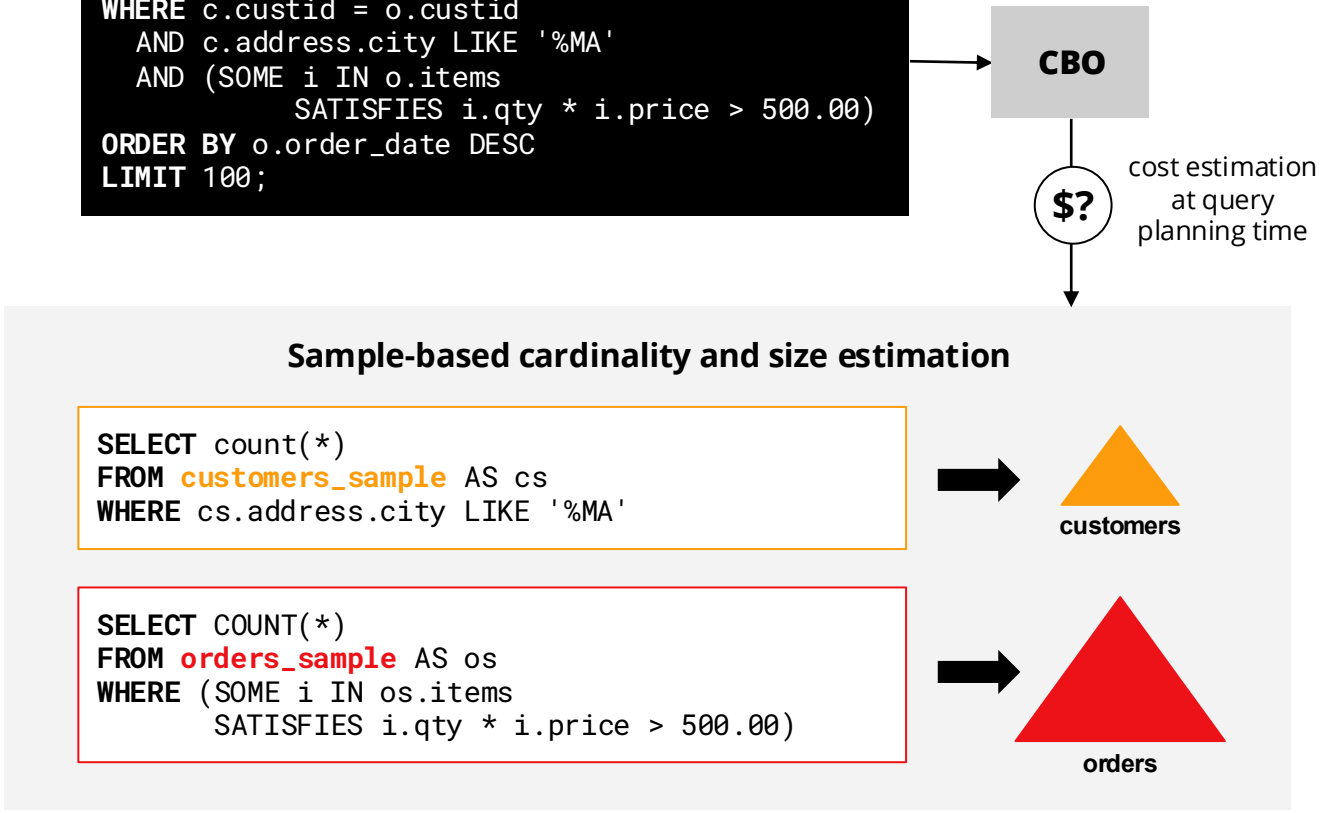
Why It Matters

- Generates the most efficient execution plan for a query
- Supports complex predicates like LIKE, expressions, functions, and correlations
- More accurate and flexible than traditional histogram-based approaches

Sampling-Based Approach

- Users periodically run ANALYZE COLLECTION to create lightweight samples of each dataset
- Couchbase Analytics uses these samples to estimate cardinality and data size
- The optimizer uses that information to make smarter decisions—like which index to use, the best join order and method, and how to group data efficiently

```
SELECT o.orderno, o.order_date, c AS cust
FROM customers AS c, orders AS o
WHERE c.custid = o.custid
  AND c.address.city LIKE '%MA'
  AND (SOME i IN o.items
           SATISFIES i.qty * i.price > 500.00)
ORDER BY o.order_date DESC
LIMIT 100;
```



Cost Based Optimization on a sample query

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Massively Parallel Processing | SQL++ Query Processing

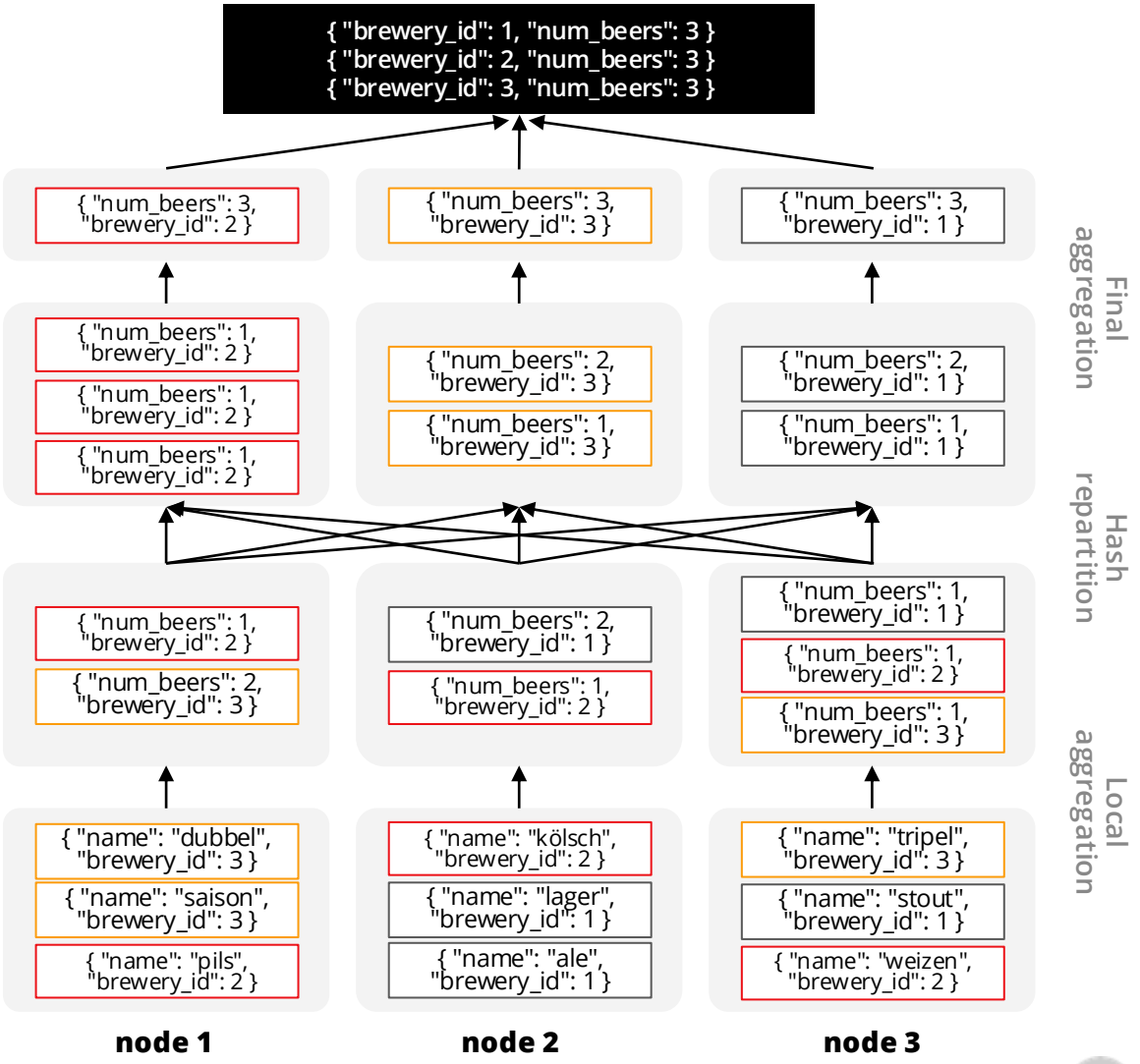
Partitioned parallelism of query processing

- Data are automatically partitioned into 128 storage partitions using each collection's primary key
- All compute units enlisted to process partitions of each query
- Each node will be responsible for processing data stored its assigned partitions
- Parallel processing of projections, selections, joins, sorting, grouped aggregation, and windowing

Relational best practices applied to JSON data

- Best of breed local algorithms spill when needed for "Big Data"
- Results in linear performance scaling for complex queries
- And with columnar storage, only "read what you need"

```
SELECT br.brewery_id, count(*) AS num_beers
FROM beers br
GROUP BY br.brewery_id;
```



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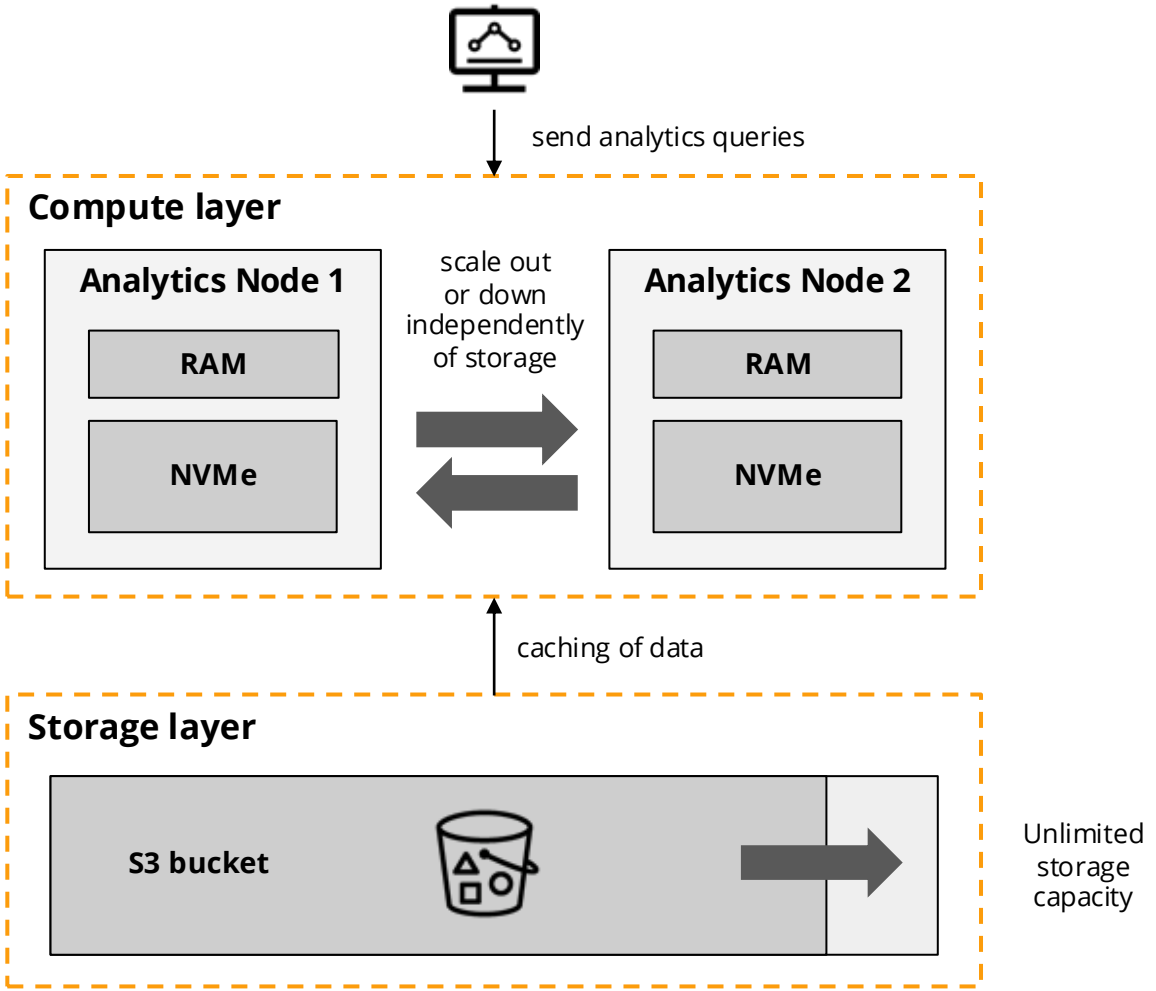
Compute and Storage Separation | Overview

Compute and Storage Architecture

- Shared-nothing compute layer with shared object storage
- Analytics clusters run on cloud compute nodes, backed by object storage
- Deployed within a dedicated, secure cloud network

Benefits

1. Compute scales independently of storage
2. Pay only for what you use—both compute and storage
3. Object storage offers virtually unlimited capacity
4. Clusters scale up or down quickly, with no need for data rebalancing
5. Linear performance scaling as compute nodes are added



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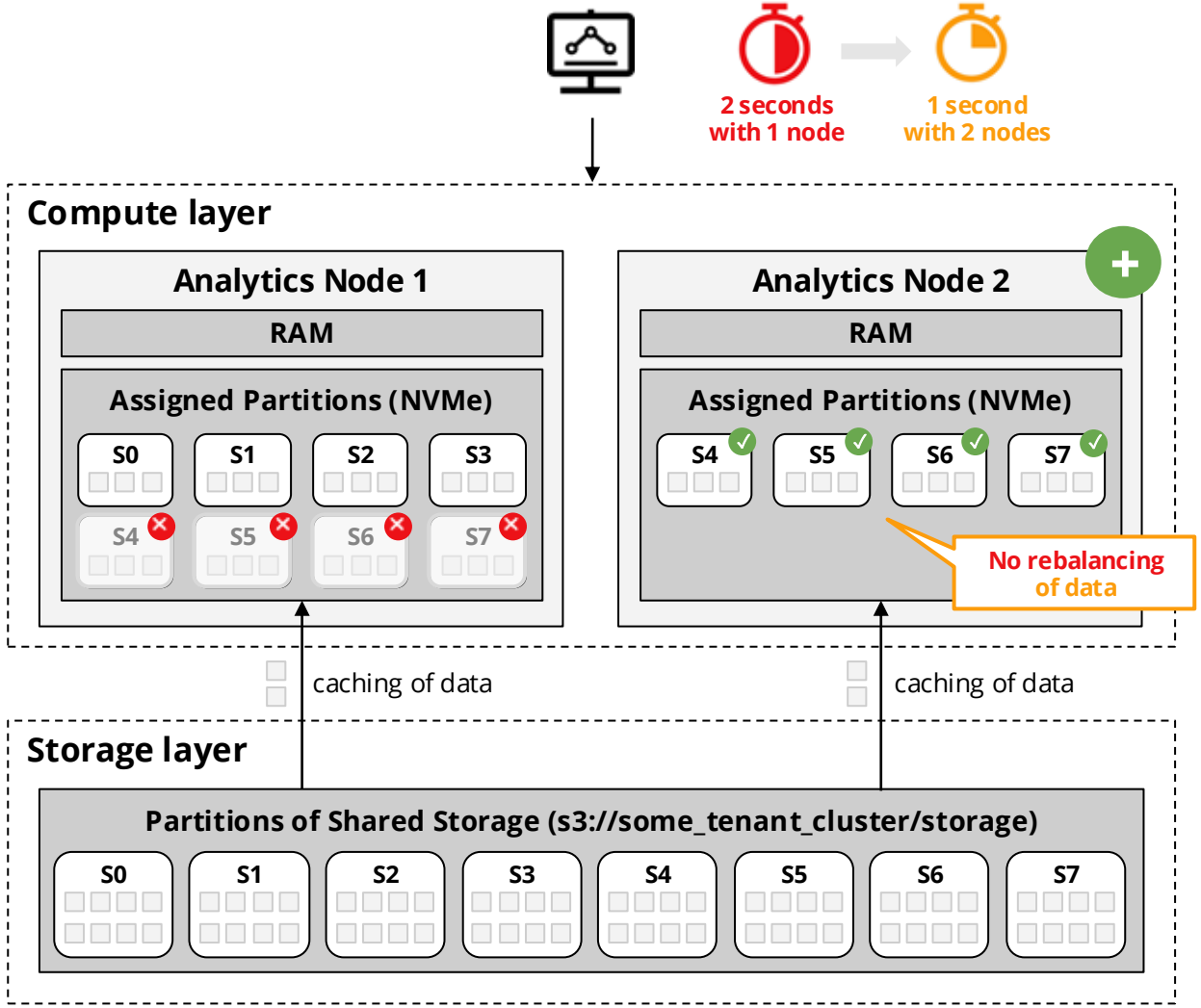
Compute and Storage Separation | Caching

Data Layers

- Local NVMe SSDs in each compute node
- Cold data stored centrally in object storage
- Active collections and indexes cached locally on compute nodes
- Cache is managed based on available space and recent usage

Partitioning and Caching Behavior

- Collections are partitioned; each compute node handles its own partitions
- On topology changes, partitions are reassigned—no data rebalancing needed
- Lazy caching: data is loaded into cache only when accessed



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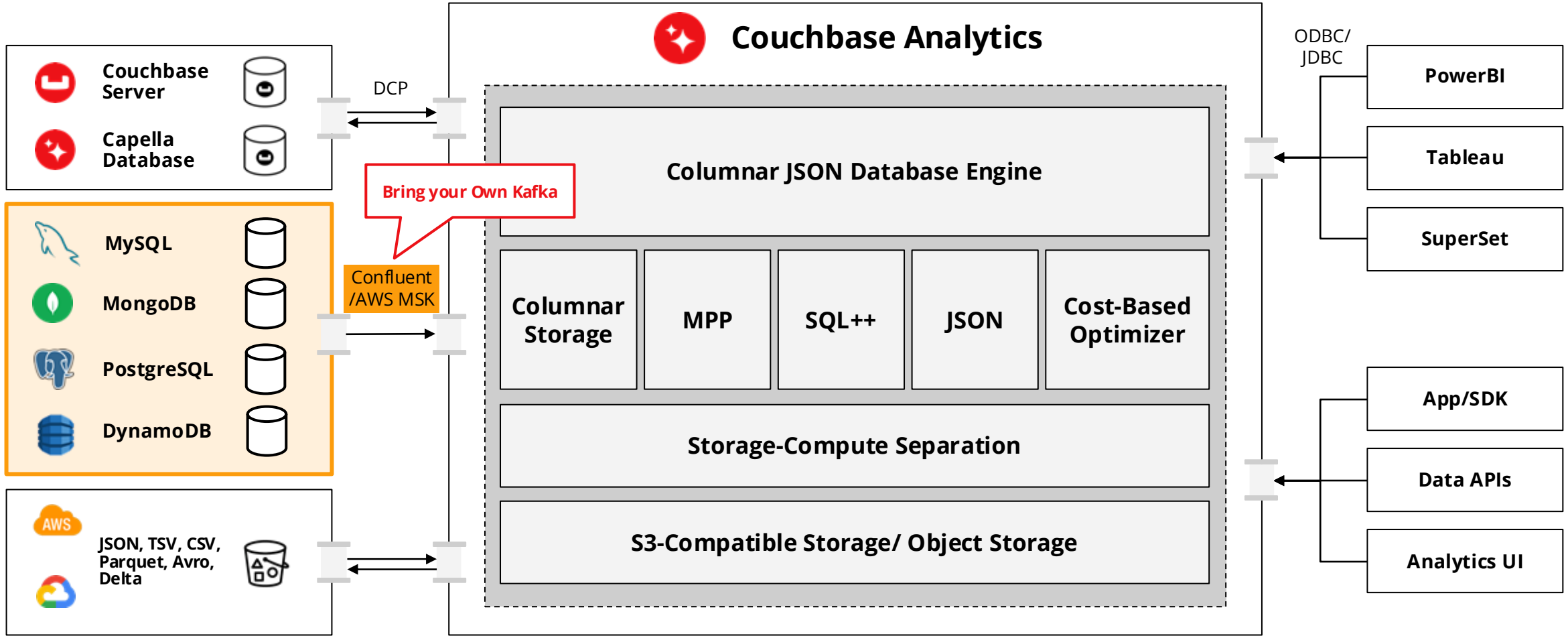
Ecosystem Integration

Deep Integration With
Multiple Sources and Destinations

Streaming Ingestion in Columnar | Bring Your Own Kafka



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Streaming Ingestion in Analytics | Kafka Link

Ingest massive volumes of data from Kafka into Analytics for a diverse set of sources **without needing a connector or writing custom code**

Support both “initial load” and **continuous (CDC) ingestion** into Analytics collections

Use UI or SDK to simplify configuration and quickly start ingesting

Ingest securely from Kafka and leverage **private connectivity** between Kafka and Couchbase Analytics

Support delivery in **Avro, Protobuf, or JSON format**

Scale out easily and **leverage built-in scaling based on Kafka partitions** to support higher throughput

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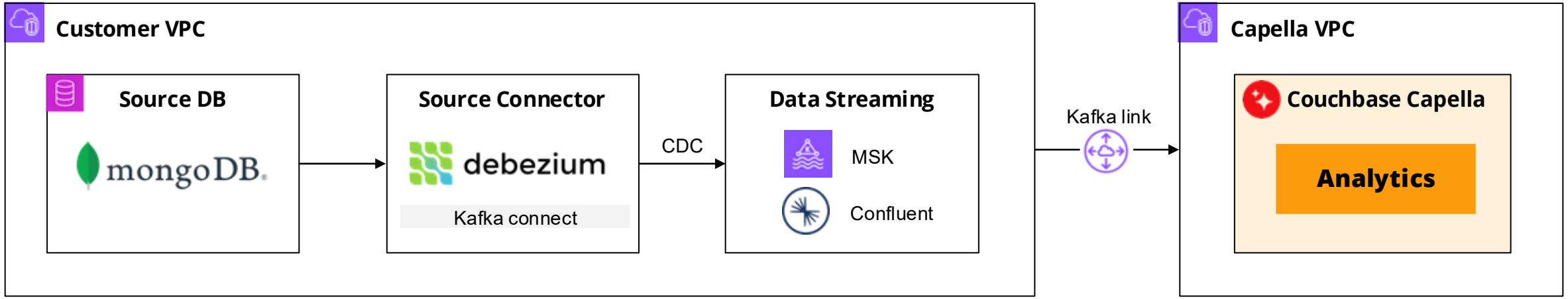
Streaming Ingestion in Analytics | Sample Setup



Data Sources	MongoDB, Postgres, MySQL, DynamoDB and several others (Coming Soon)
Kafka platforms	MSK, Confluent
Kafka authentication	IAM, OAuth, Plain, SHA-SCRAM
Kafka schema registry	OSS Schema registry
Kafka private connectivity	VPC peering, Private Link

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External Collections | Interact with External Object Storage

External collections

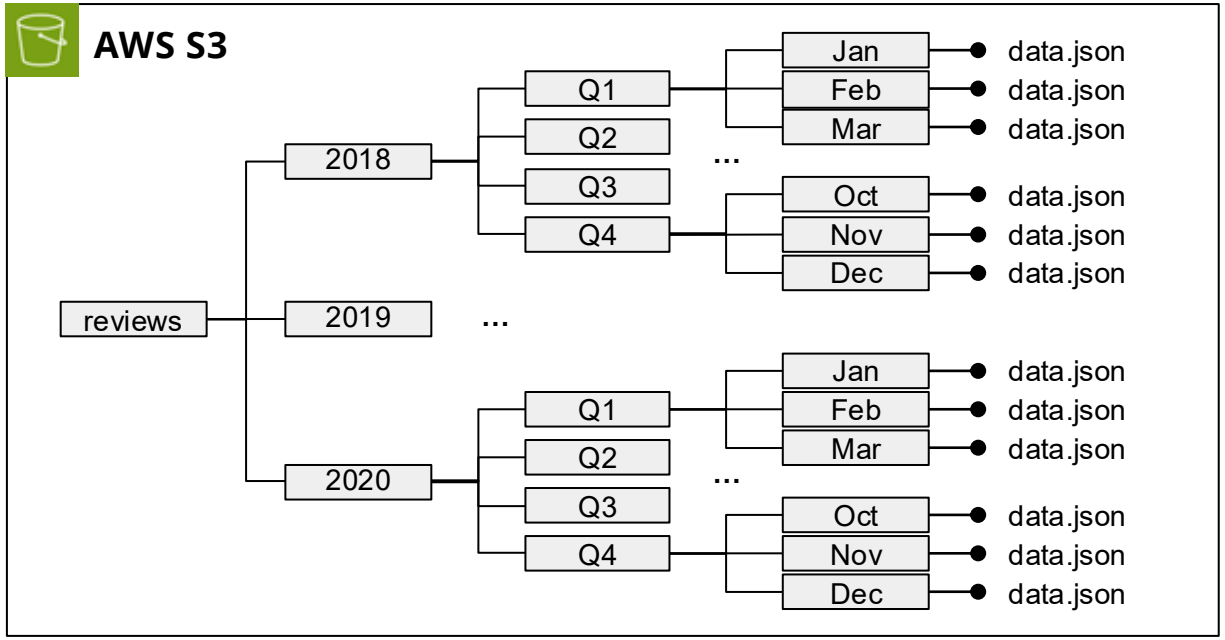
- An external collection makes S3 or GCS (or on-prem S3) data look like an internal collection
- Multiple file formats supported: JSON, CSV/TSV, Parquet, Avro and Delta

DML options to access External collections

1. CREATE COLLECTION AS (query) statement, when querying external data
2. COPY TO statement to write data to a structured external collection
3. COPY FROM statement to load from S3 or GCS (or on-prem S3) into a standalone collection

Dynamic paths for querying external data

- Paths and filters available to indicate which folders to include/exclude
- Dynamic paths turn folders into queryable data



Couchbase Analytics

```
CREATE EXTERNAL COLLECTION myExternalCollection
ON `myReviewsContainer`
AT myExternalLink
PATH "reviews/{year: int}/{quarter: string}/{month: string}"
WITH {"format": "json"};
```

Dynamic prefixes

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External Collections | Querying an External Object Storage (S3)

```
SELECT * FROM myExternalCollection m
WHERE year = 2019
AND quarter = "Q1" OR quarter = "Q4";
```

query

metadata

AWS S3

read data in parallel

Capella Analytics

Query Compilation

Retrieve metadata from external storage

reviews/2019/Q1/Jan/data.json
reviews/2019/Q1/Feb/data.json
reviews/2019/Q1/Mar/data.json
reviews/2019/Q2/Apr/data.json
reviews/2019/Q2/May/data.json
reviews/2019/Q2/Jun/data.json
reviews/2019/Q3/Jul/data.json
reviews/2019/Q3/Aug/data.json
reviews/2019/Q3/Sep/data.json
reviews/2019/Q4/Oct/data.json
reviews/2019/Q4/Nov/data.json
reviews/2019/Q4/Dec/data.json

Apply Filters

reviews/2019/Q1/Jan/data.json
reviews/2019/Q1/Feb/data.json
reviews/2019/Q1/Mar/data.json
reviews/2019/Q4/Oct/data.json
reviews/2019/Q4/Nov/data.json
reviews/2019/Q4/Dec/data.json

Assign files to the partitions on all nodes

partition P1
reviews/2019/Q1/Jan/data.json
reviews/2019/Q4/Dec/data.json

partition P2
reviews/2019/Q4/Oct/data.json
reviews/2019/Q4/Nov/data.json

partition P3
reviews/2019/Q1/Feb/data.json
reviews/2019/Q1/Mar/data.json

Read data from all partitions in parallel

partition P1
{ ,,, }, { ... }, { ... }, ...

partition P2
{ ,,, }, { ... }, { ... }, ...

partition P3
{ ,,, }, { ... }, { ... }, ...

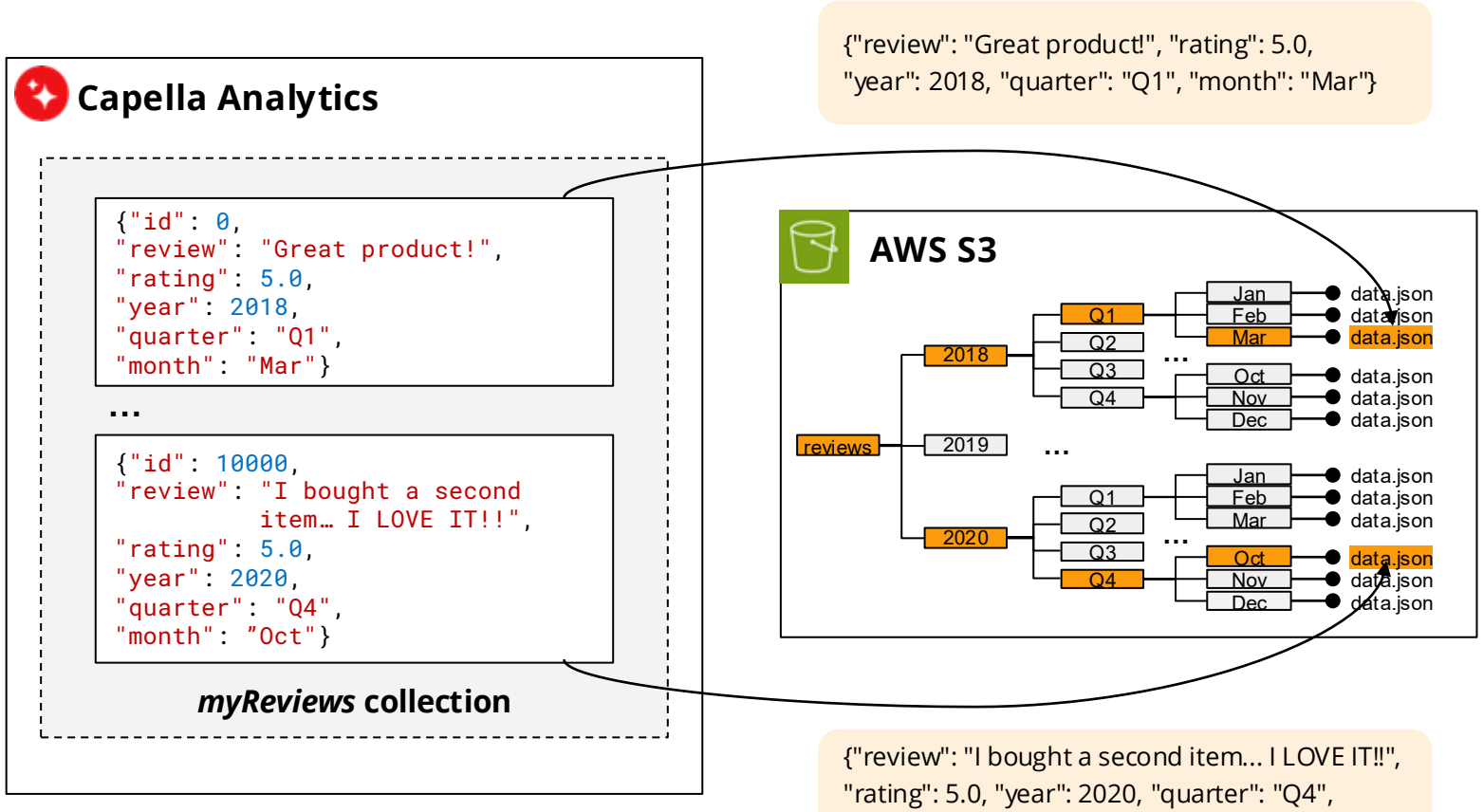
```
PATH "reviews/{year: int}/{quarter: string}/{month: string}"
```

Dynamic prefix from the External collection

COPY TO | External Storage (e.g. AWS S3)

- COPY TO statement to write data from an Analytics collection to a structured External collection

```
COPY (
   SELECT review, rating, year,
          quarter, month
   FROM myReviews
) r
TO `myReviewsContainer`
AT `myExternalLink`
PATH("reviews", year, quarter, month)
OVER(
   PARTITION BY r.year year,
                r.quarter quarter,
                r.quarter month
)
WITH {
   "format": "json"
}
```



{"review": "Great product!", "rating": 5.0, "year": 2018, "quarter": "Q1", "month": "Mar"}

{"review": "I bought a second item... I LOVE IT!!", "rating": 5.0, "year": 2020, "quarter": "Q4", "month": "Oct"}

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COPY TO | Couchbase Operational Database

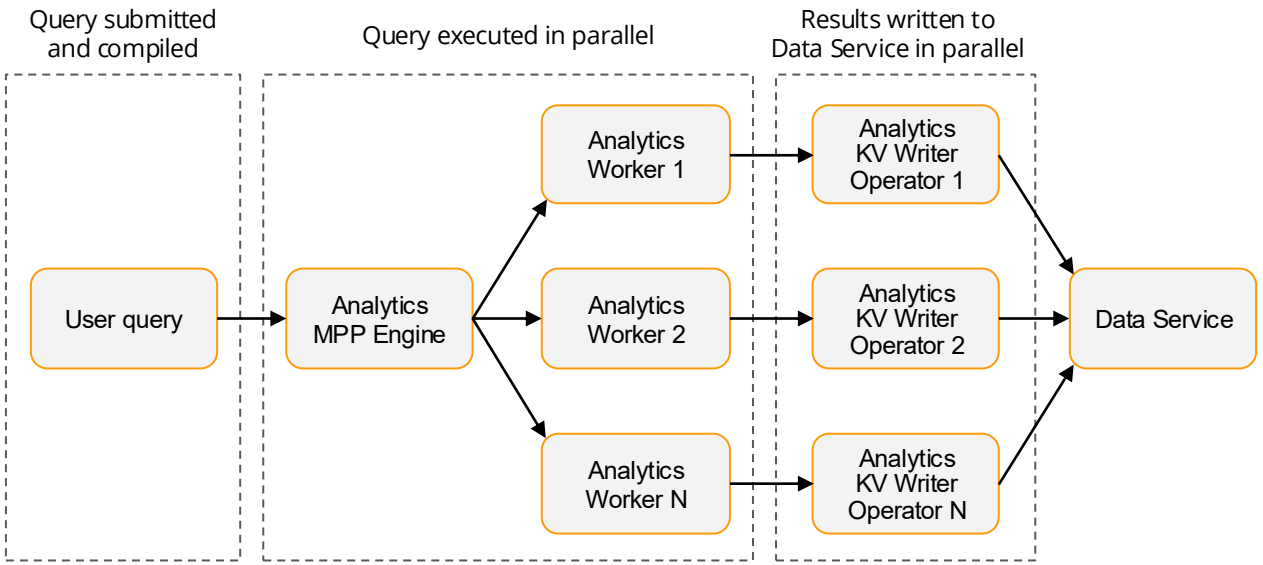
```
COPY (
    SELECT VALUE c
    FROM Commerce.Customers c
    WHERE c.city IN ["Boston, MA", "Hanover, MA"]
  ) AS t
TO myBucket.myScope.CustomerDetails
AT myCouchbaseRemoteLink
KEY AUTOGENERATED
```

Benefits

- Move insights natively from Analytics → Operational DB.
- No reverse ETL or custom apps needed.

How it works

- Run any Analytics query (joins, views, UDFs, etc.).
- Persist results into a Couchbase Data Service collection using the COPY (query) TO <targetCollection> command.
- Executes as a batch operation.



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BI Tools | Support and Tabular Analytics Views

Support for Tableau, PowerBI and SuperSet

- Data analysts can use their favorite tools
- JDBC/ODBC connectivity "under the hood"
- Power BI: Basic, Client certificate and LDAP auth
- Tableau, Superset: Basic, Server certificate auth

Tabular Analytics Views (TAV) of non-tabular data

- Defined using SQL++ queries, such as UNNEST nested data
- View "signatures" provide the mappings to SQL data types
- View constraints provide the metadata - the primary/foreign keys
- Underlying data is still document data: views are not materialized

customers

```
{ "custid": "C13",
  "name": "T. Cruise",
  "address": { "street": "201 Main St.","city": "St. Louis, MO", "zipcode": "63101" },
  "rating": 750 }
```

TAV

custid	name	rating	street	city	zipcode
C13	T. Cruise	750	201 Main St.	St. Louis, MO	63101
...	...	...	...	...	...

orders

```
{ "orderno": 1002,
  "custid": "C13",
  "order_date": "2017-05-01",
  "ship_date": "2017-05-03",
  "items": [ { "itemno": 460, "qty": 95,"price": 100.99},
             { "itemno": 680, "qty": 150, "price": 8.75} ]
}
```

TAV

orderno	custid	order_date	ship_date	total
1002	C13	2017-05-01	2017-05-03	10906.55
...	...	...	...	...

items

TAV

orderid	itemno	itemqty	item_price
1002	460	95	100.99
1002	680	150	8.75
...	...	...	…

products

```
{ "itemno": 680,
  "category": "essentials",
  "name": "Beer Opener",
  "description": "Robotic beer bottle opener",
  "manuf": "Robo Brew",
  "listprice": 29.95
}
```

TAV

itemno	category	name	description	manuf	listprice
680	essentials	Beer Opener	Robotic ...	Robo Brew	29.95
...	...	...	...	...	...

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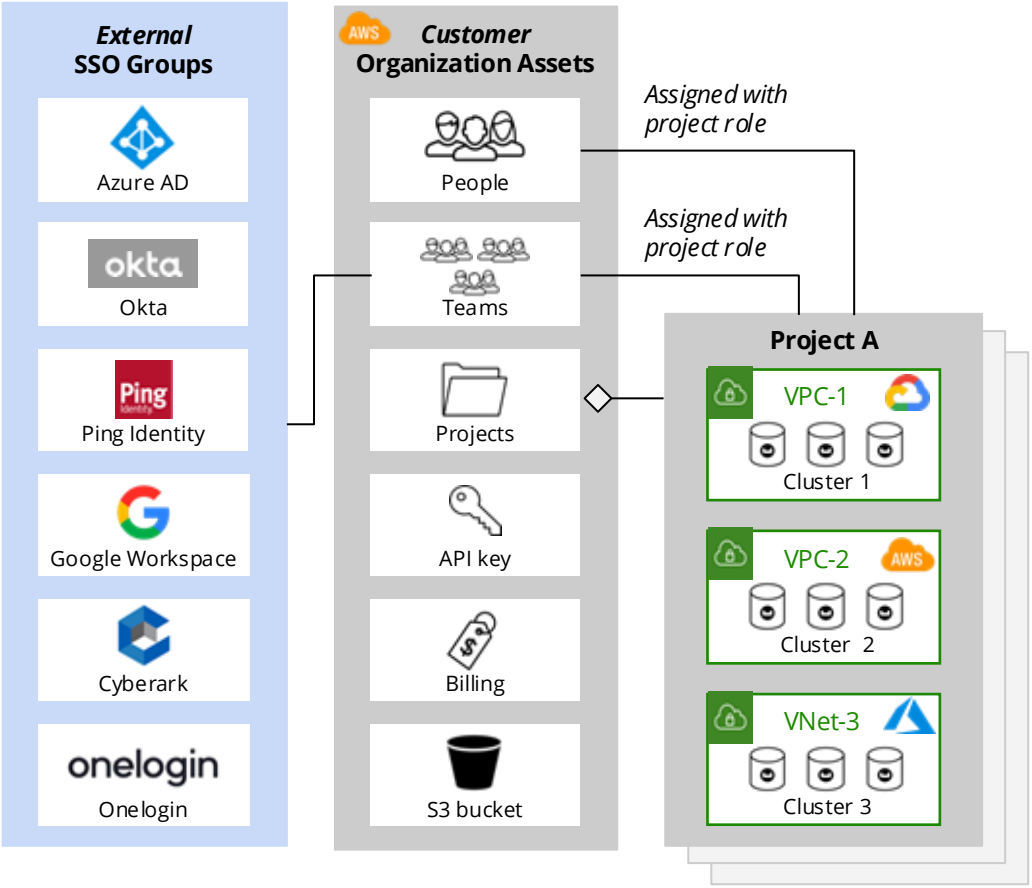
Management & Operations

Organize and Manage
Databases, Teams and Projects

Management | Users and Projects



Analytics Hierarchy

- Organization > Projects > Clusters > Databases > Scopes > Collections

Organization

- An Organization is laid out as a Hierarchy
- Organization members do not have data access by default

Projects

- Projects are used to organize groups of clusters
- People must be added to projects to access clusters
- You can also add SSO groups (Teams) with project role

Cluster

- Clusters are group of one or multiple compute nodes

Database/Scope/Collection

- A database is the top-level container for organizing data
- Scopes are group of related collections
- Collections are queryable and manipulable containers for data access

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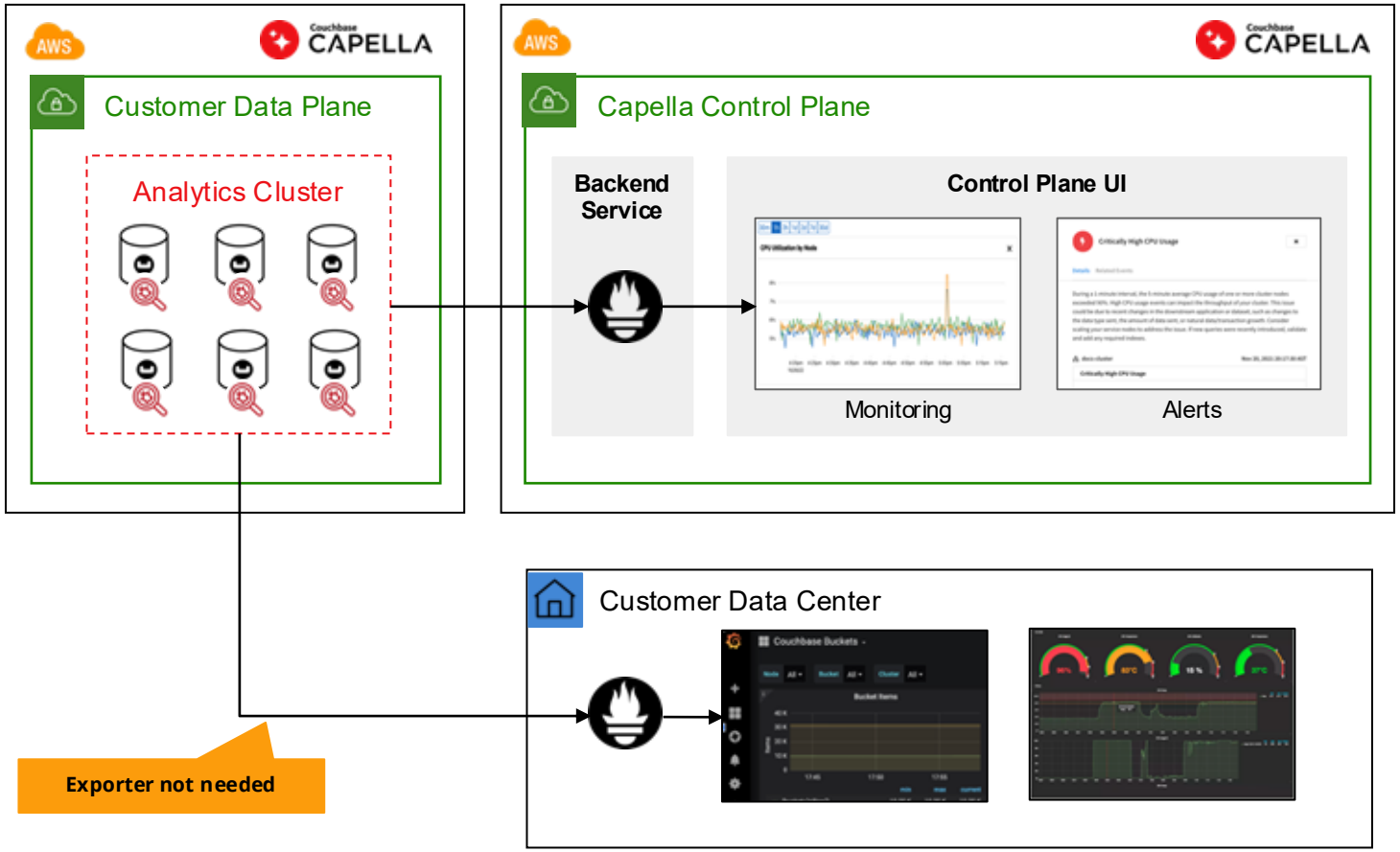
Management | Monitoring and Alerting



Monitoring

- Capella has its own highly available Prometheus infrastructure
- Curated set of metrics obtained from the Analytics Databases
- Configurable UI

Alerts

- Set of conditions which generate Alerts
- Alerts are actionable - suggested Action
- Alerts notified via email and logged in the Activity Log as well *(subscribe!)*

Self-managed Prometheus

- No need of exporter
- Both /metrics and the federated /prometheus endpoints available
- Full set of metrics

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Management | Monitoring UI

Node Level

- I/O: Read/Write rate
- Resource usage: CPU, Memory and System load

Node & Instance level

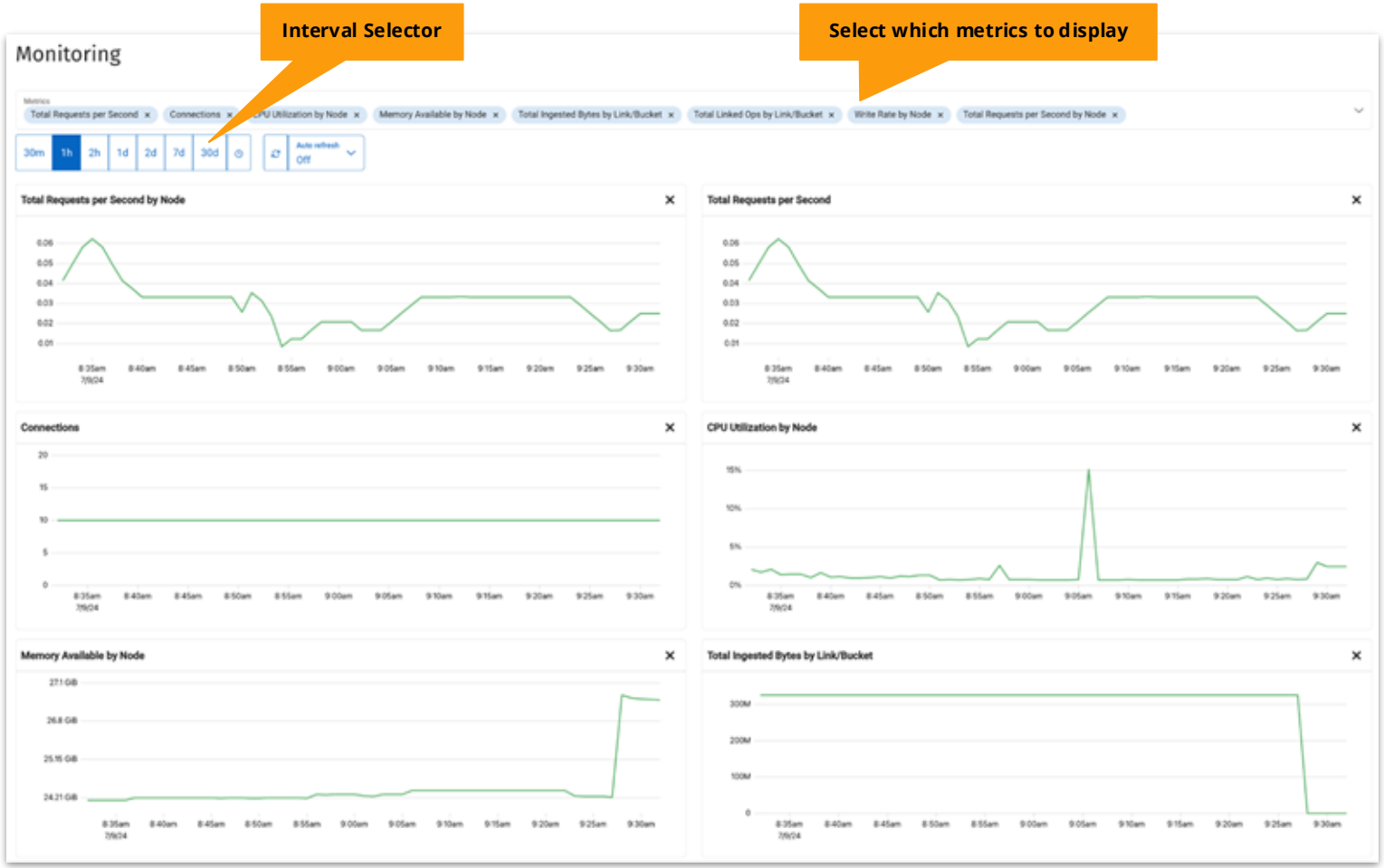
- Total request per second

Self-Instance level

- # of connections

Bucket/Link & Instance level

- Ingested bytes rate
- Linked ops rate
- Parse failure rate



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Management | Query Performance & Analysis

SQL++ functions

- `completed_requests()` and `active_requests()`
- 1k request history is available for analysis

Query Analysis

Find the top 5 slowest queries

```
SELECT VALUE r FROM completed_requests() AS r
ORDER BY r.elapsedTime DESC LIMIT 5;
```

Queries taking > 5 minutes

```
SELECT VALUE r FROM completed_requests() AS r
WHERE r.elapsedTime / 60 > 5;
```

Queries taking > 1GB of Memory

```
SELECT VALUE r FROM completed_requests() AS r
WHERE r.jobRequiredMemory > 1000000000;
```


```
{
  "cancellable": true,
  "clientContextID": "b58848c3-b267-4711-90a2-2ff1dd300e4a",
  "elapsedTime": 0.022,
  "jobCreateTime": "2024-04-25T20:09:36.389",
  "jobEndTime": "2024-04-25T20:09:36.392",
  "jobId": "JID:0.13",
  "jobQueueTime": 0,
  "jobRequiredCPUs": 2,
  "jobRequiredMemory": 524288,
  "jobStartTime": "2024-04-25T20:09:36.389",
  "jobStatus": "TERMINATED",
  "node": "127.0.0.1:9600",
  "plan": "",
  "remoteAddr": "127.0.0.1:55857",
  "requestTime": "2024-04-25T20:09:36.370",
  "scanConsistency": "not_bounded",
  "state": "completed",
  "statement": "SELECT VALUE al FROM airline al WHERE al.name = '40-Mile Air';",
  "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
        AppleWebKit/605.1.15 (KHTML, like Gecko) Version/17.2.1 Safari/605.1.15",
  "users": "couchbase",
  "uuid": "2e60af04-9bbc-4876-a2ff-0eebacca6b58"
}
```


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Management | Activity Log

Activity Log Overview

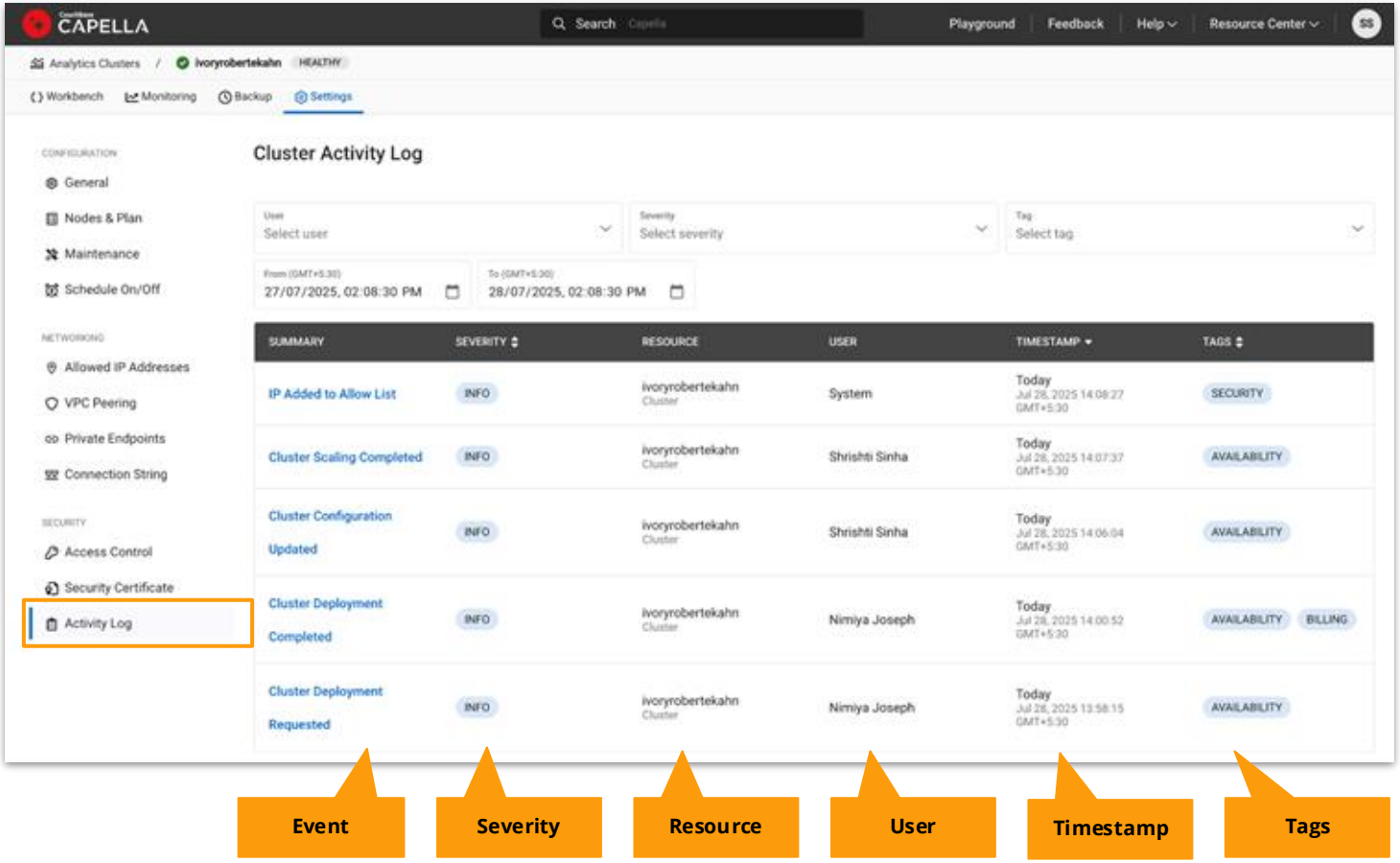
- Audit trail of important events
- Shows who, when and what
- Available at the Organization, Project or Cluster level

Type of Events

- Cluster deployment
- Cluster turned On/Off
- Backup and Restore
- Maintenance
- API Keys
- and more

Alerts logging

- Stores the alerts raised by the system



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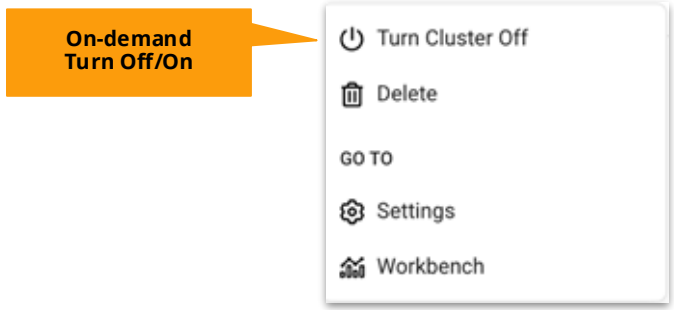
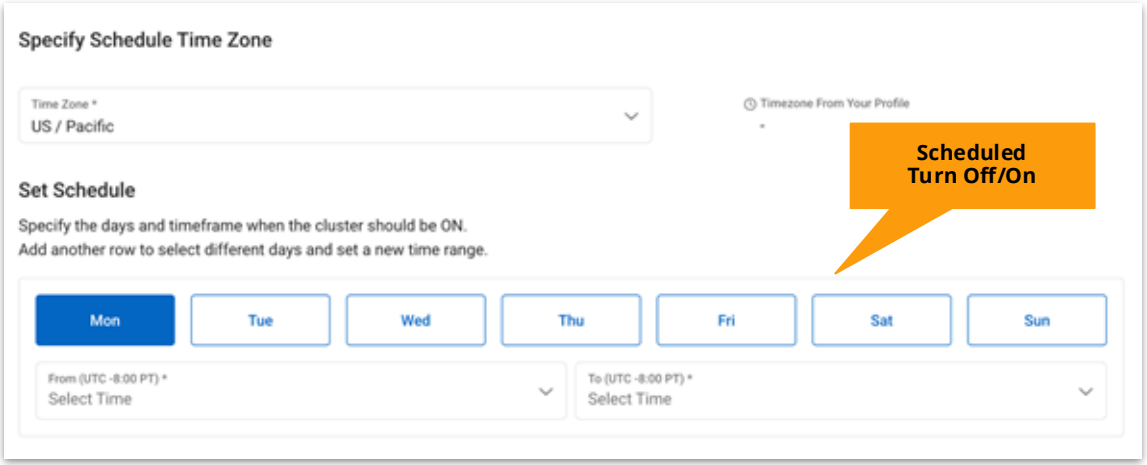
Management | Turning Cluster On/Off





Analytics Clusters can be turned off and turned on

- On-demand
- Using a schedule to choose days of week and times
- Using management API's (v4) -- schedule, turn on, turn off

When Turned Off (no compute)

- Only charged for storage
- All connections are terminated

On-demand Turn Off and On

- Can be turned off for a maximum of 30 days
- Automatically turned on after 30 days to keep the cluster up to date with the latest maintenance patches
- Notification is sent 2 days before it auto turns on

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Operations | Scaling

Horizontal compute scaling

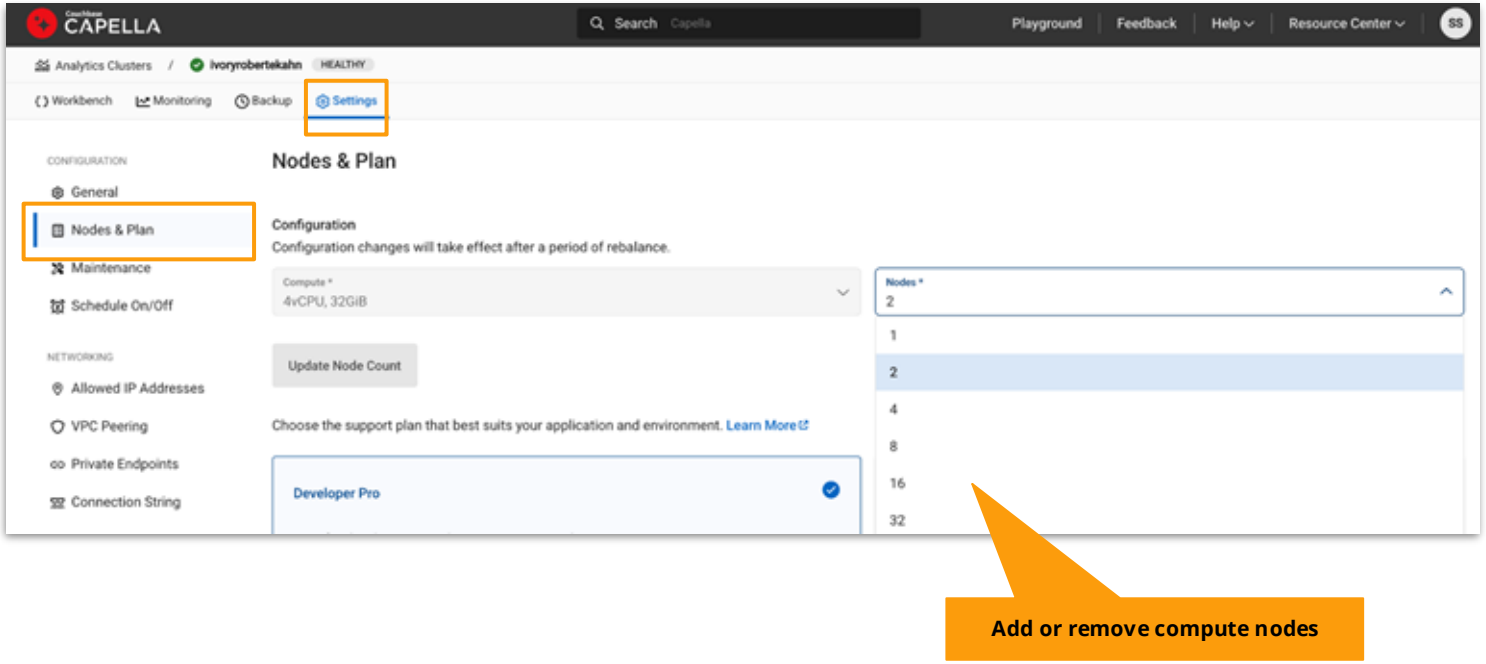
- Changes the number of nodes

Vertical compute scaling

- Changes the CPU/RAM per Node
- NOT ALLOWED after cluster creation

Storage scaling

- Automatically scaled up and down based on the usage
- Charged only on used capacity



Add or remove compute nodes

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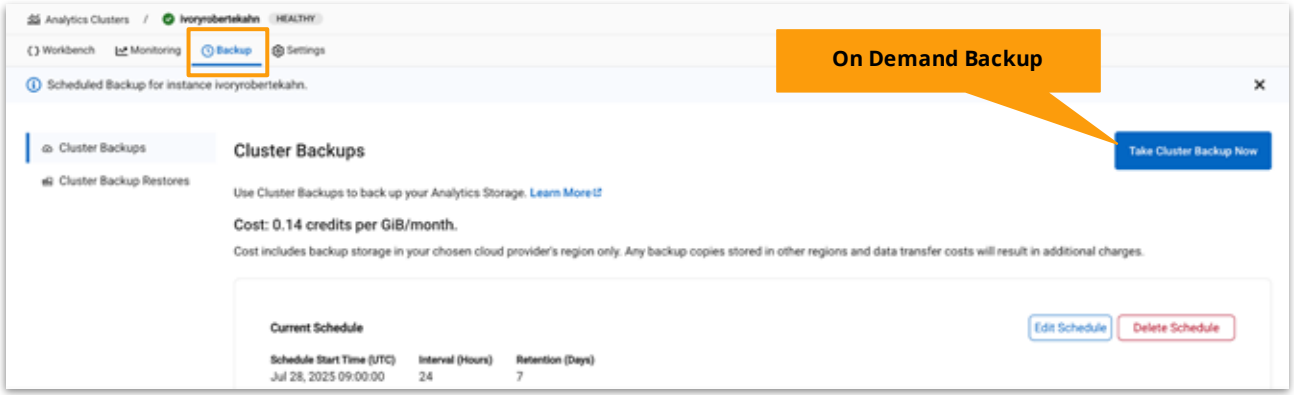
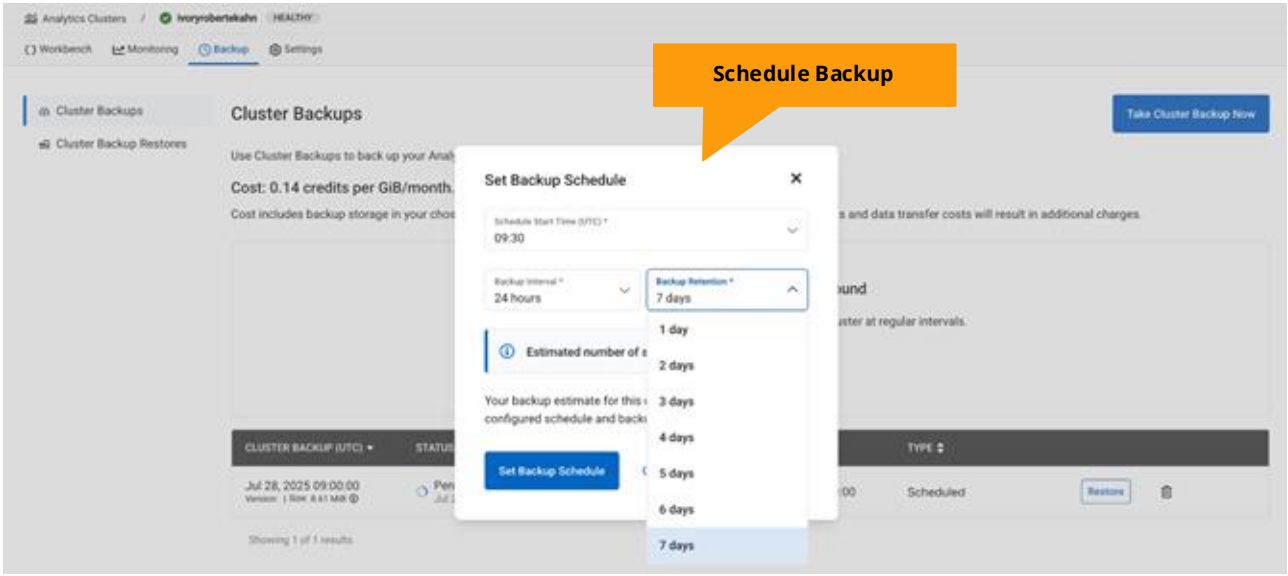
Operations | Backup and Restore





Backup & Restore Use Cases

- As part of an overall disaster recovery plan for production data, a robust scheduled backup and retention policy is recommended
- Only on AWS

Backup

- Uses AWS S3 Backup Service
- Per Capella Analytics Cluster
- On Demand or Scheduled on Weekly cycles
- Backup Interval 1-24 Hours ; Retention : 1-30 days
- Backup of Data, Metadata and Links

Restore

- To the same Cluster on the same CSP
- Easy reconnection on restore
- IP allowed lists are not restored for security reasons

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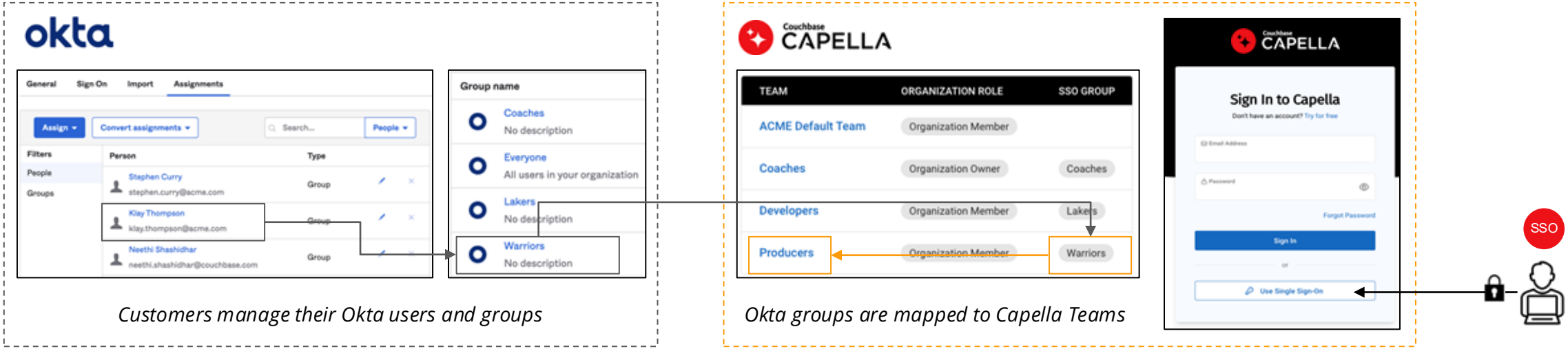
Security

Multi-layered, End-to-End Protection

Security | Single Sign-On (SSO)



Single Sign-On (SSO)

- Azure AD, Okta, Ping Identity, Google Workspace, CyberArk and OneLogin supported
- Users make use of their existing corporate credentials
- Authentication delegated to the SSO Provider

SSO Groups mapped to Capella Teams

- SSO must be enabled in Capella (at the Org level)
- Capella Teams are given Organization Role
- Capella Teams are assigned to Projects with Project Roles
- Users can click on the SSO button to login

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Security | Role-Based Access Control (RBAC)

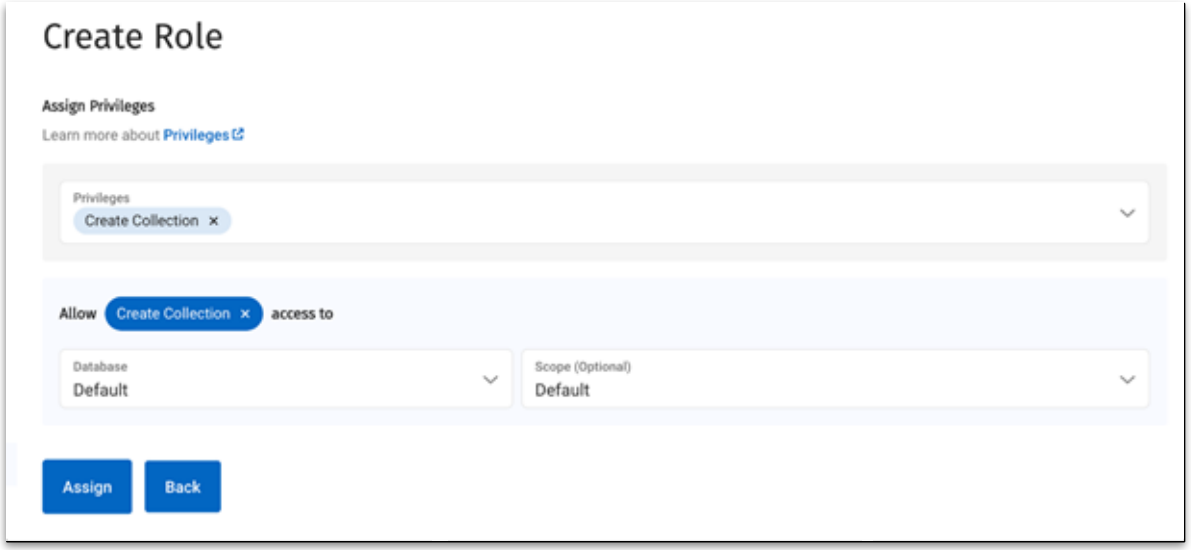
Purpose

- Allows the creation of new roles and assignment of privileges to these roles
- Roles can be assigned to accounts
- Privileges can be directly added to accounts as well

Benefits

- **Enhanced Security**: Fine-grained control over access
- **Scalability**: Easily manage access as the number of users grows
- **Flexibility**: Customize roles and privileges to fit organizational requirements

Key Components	
Role creation	Define custom roles based on specific needs
Privilege Assignment	Assign specific privileges to each role
Account Association	Link one or more roles to accounts
Direct Privileges	Option to directly assign privileges to accounts without using roles or in addition to



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Security | Compliance

GDPR

Analytics is designed to meet global data privacy requirements and enable customers' compliance with the GDPR

SOC II

Analytics complies with our SOC II controls, audit/certification process pending

HIPAA

Analytics supports HIPAA-regulated applications, audit/certification process pending

PCI DSS

Analytics is PCI DSS compliant, certification process pending

CSA STAR

Analytics complies with our CSA STAR controls, audit/certification process pending

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Thank you!

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